MedLens: Sprint 2 Project Planning

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1 PROJECT TOPIC

1.1 Background

When you are prescribed medication as a patient, the main thing you and your doctor want to know is if that medication helped move your problem in the correct direction. For example, if a patient has high cholesterol, and the doctor prescribed tablets to be taken, we want to know if those tablets had an effect on lowering the patients cholesterol levels.

Getting an answer to these types of questions is high friction. In order for someone to analyze this interaction, they would have to tediously check back and forth between the different Electronic Health Records (EHR) systems. They would have to check their prescribed medications in one system, check their lab results in another and then try to cross-reference information. Due to all these different systems, there are many cases of doctors failing to prescribe a supplemental medication to counteract the side-effects of the primary medication.

Our app MedLens is meant to help bridge the gap between these different systems. MedLens is meant to give doctors and patients a more clear visual of the changes in a patient's laboratory results versus their medication history. This also allows us to bridge a gap in safety and flag whenever a supplemental medication order is missing.

1.2 Justification

Tracking medication changes and effectiveness is an ongoing problem for both Clinicians and Patients. Clinicians want a simple way to review charts and make insightful deductions about their patients' health, while patients deserve a concise way to track their health that is easy to understand for them.

About half of the adults in the US have high blood pressure, however only a quarter of those adults have their high blood pressure under control. About 70% of patients that have high blood pressure are already on some type of

medication, and knowing if the medication is working is important to them with rising drug and medical care costs. The cost of high blood pressure related treatments crossed $219 billion annually in the United States in 2019 according to the CDC.

Current dashboards and charts display medications and lab results but do not track health for specific conditions in a holistic manner over time. A simple tool that consolidates FHIR data into more informative and meaningful health tracking would save clinicians time reviewing patient charts and give patients a transparent picture of their own health and treatments in an easy to understand way.

1.3 Solution

MedLens is a web application that connects to the FHIR R4 server and pulls in a single patient's medical history. With that data, insightful conclusions can be made in the form of:

- **Timeline**: Medical data drawn over a period of time indicating trends over time in lab results, with each metrics goal range highlighted.
- **Scorecard**: A summary of how each medication results in changes in specific lab results with emphasis on specific goal ranges and an assessment of medication to results relation in the form verification of the type of trend.
- **Safety Panel**: A checklist of outlined risks, warnings, and todos for the patient. This will be sourced from scheduled lab visits, known drug pairings, and current medication use tracking (limited to 5 ongoing medications).
- **Comparison**: This patient's data plotted against other patients on the same medication. With data privacy in mind, the application will compare the patients current labs and prescribed medication to other people with the same or similar condition. For a generalized view of if a medication is working with what side effects.

This solution will focus specifically on blood pressure, A1c, glucose, potassium, creatinine, LDL, HDL, and weight in order to give holistic overviews about Cholesterol, High Blood Pressure, and Diabetes.

The application will be automatically directed to the views that matter the most to them. For a clinician, they will see specific codes and statistics for multiple patients with granular data filtering over specific patients. For the patient, they will see their medical history in plain language with technicalities abstracted away, and graph views of their health progress. They will be able to see if a certain type of medication is causing unintended changes in lab results while trying to achieve their health goals.

This app will be read-only, and purely tracking data that is gathered from verified sources such as prescribed meditation and any necessary lab results that a clinician orders for their patient.

2 TECHNICAL DESIGN

2.1 Tools/Technology

Area	**Choice**
Language	Python 3.11
UI framework	Streamlit
FHIR access	requests (HTTP library) against the FHIR R4 REST API
Data handling	pandas
Charts	Plotly / Matplotlib / Seaborn
Statistics	statsmodels, SciPy
Clinical rules	CSV files with medication interactions, monitoring, targets and goal ranges sourced from reputable separate data sources
Hosting	Streamlit Community Cloud
Source control	GitHub

Database or backend service is not required, since all data is read directly from FHIR.

Datasets and Data Sources

- **SMART R4:** (https://r4.smarthealthit.org): an open FHIR server with preloaded data and no authentication needed. This is the primary source.
- **Synthea**: Test data generation that will be loaded locally through a Docker container. Fallback if SMART Health does not have the data we need.
- **Resources read:** Patient, MedicationRequest, Observation and Condition.
- **Rule tables:** about 15 to 20 medications relating to clinical rules for Cholesterol, Diabetes and High Blood Pressure. Sourced from reputable separate data sources

2.2 Architecture Diagram

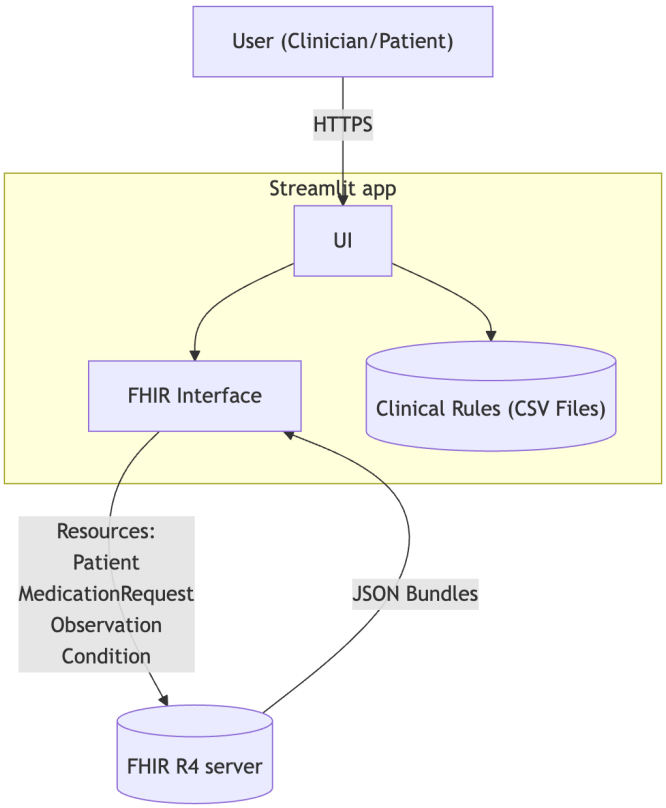


Figure 1—Architecture diagram for MedLens Application.

2.3 Screen Mock-up

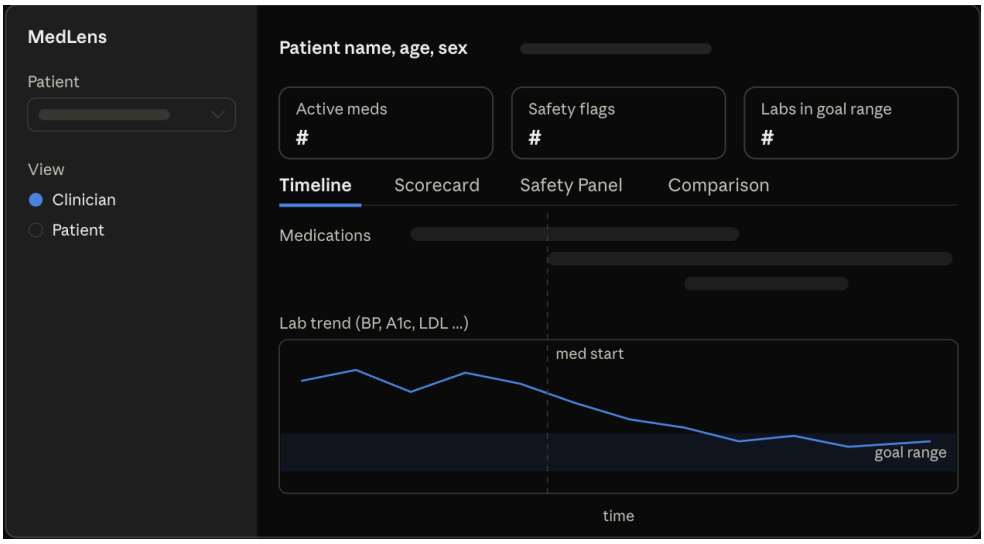


Figure 2—Screen Mock-up for MedLens Application

3 IMPLEMENTATION PLAN

Week	Dates	Task	Needs / Risks
1	Oct 26 - Nov 1	• Sprint 2: finalize planning document with mentor. • Set up GitHub repo, Streamlit and Community Cloud deployment. • Build a client to interface with the FHIR system. Develop sidebar and header components.	• Need: mentor feedback on scope before Week 2. • Risk: Community Cloud outbound access to the FHIR server; fallback is local demo.
2	Nov 2 - Nov 8	• Implement FHIR requests to retrieve and parse medication period and observation data. • Build timeline view with goal-range bands and test with demo patients.	• Risk: few patients with sparse medical history; may need to generate test data.

Week	Dates	Task	Needs / Risks
3	Nov 9 - Nov 15	• Sprint 3 check-in • Build the clinical rules (CSV). • Implement before/after and range calculations with verdict badges. • Implement regression analysis and response charts based on before/after periods. • Record Demo Video	• Need: credible public sources for clinical rules. • Need: Verify demo video with mentor
4	Nov 16 - Nov 22	• Import clinical rules csv and apply checks from the csv into data model for safety panel and score card. • Implement monitoring for concurrent medication use.	
5	Nov 23 - Nov 29	• Sprint 4 check-in • Implement patient group fetch by medication. • Build patient group comparison charts and add generalized information markers based on established ranges. • Implement clinician/patient toggle to switch views.	• Risk: datasource may lack enough patients per drug; generate test patient groups.
6	Nov 30 - Dec 6	• Sprint 5 / final: report, final deployment.	• Need: final deliverable requirements confirmed with mentor

4 REFERENCES

1. Hurley, Judith S et al. “Laboratory safety monitoring of chronic medications in ambulatory care settings.” Journal of general internal medicine vol. 20,4 (2005): 331-3. doi:10.1111/j.1525-1497.2005.40182.x
2. “High Blood Pressure Facts.” Centers for Disease Control and Prevention, Centers for Disease Control and Prevention, www.cdc.gov/high-blood-pressure/data-research/facts-stats/. Accessed 22 Oct. 2026.